

Contents

1	README	2
2	Chat.java	6
3	ChatterBot.java	7

1 README

```
1  eladfirer_200
2  212155584
3
4  say something
5  Don: say something? okay: something
6  Julio: say something? okay: something, bon appetit
7  Don: something? okay: something, bon appetit something? okay: something, bon appetit, are you satisfied?
8  Julio: please let me know what to say, i dont understand what is something? okay: something,
9  bon appetit something?
10 okay: something, bon appetit, are you satisfied?
11 Don: say i should say please let me know what to say, i dont understand what is something? okay: something,
12 bon appetit
13 something? okay: something, bon appetit, are you satisfied?
14 Julio: say i should say please let me know what to say, i dont understand what is something? okay: something,
15 bon
16 appetit something? okay: something, bon appetit, are you satisfied?, bon appetit
17 Don: say i should say please let me know what to say, i dont understand what is something? okay: something,
18 bon appetit
19 something? okay: something, bon appetit, are you satisfied?, bon appetit? okay: i should say please let me
20 know what to
21 say, i dont understand what is something? okay: something, bon appetit something? okay: something, bon
22 appetit, are you
23 satisfied?, bon appetit
24 Julio: say i should say please let me know what to say, i dont understand what is something? okay: something,
25 bon
26 appetit something? okay: something, bon appetit, are you satisfied?, bon appetit? okay: i should say please
27 let me know
28 what to say, i dont understand what is something? okay: something, bon appetit something? okay: something, bon
29 appetit,
30 are you satisfied?, bon appetit, bon appetit
31 Don: say i should say please let me know what to say, i dont understand what is something? okay: something,
32 bon appetit
33 something? okay: something, bon appetit, are you satisfied?, bon appetit? okay: i should say please let me
34 know what to
35 say, i dont understand what is something? okay: something, bon appetit something? okay: something, bon
36 appetit, are you
37 satisfied?, bon appetit, bon appetit ? stop telling me to say i should say please let me know what to say, i
38 dont
39 understand what is something? okay: something, bon appetit something? okay: something, bon appetit, are you
40 satisfied?,
41 bon appetit? okay: i should say please let me know what to say, i dont understand what is something? okay:
42 something,
43 bon appetit something? okay: something, bon appetit, are you satisfied?, bon appetit, bon appetit
44 Julio: i like saying i should say please let me know what to say, i dont understand what is something? okay:
45 something,
46 bon appetit something? okay: something, bon appetit, are you satisfied?, bon appetit? okay: i should say
47 please let me
48 know what to say, i dont understand what is something? okay: something, bon appetit something? okay:
49 something, bon
50 appetit, are you satisfied?, bon appetit, bon appetit ? stop telling me to say i should say please let me know
51 what to
52 say, i dont understand what is something? okay: something, bon appetit something? okay: something, bon
53 appetit, are you
54 satisfied?, bon appetit? okay: i should say please let me know what to say, i dont understand what is
55 something? okay:
56 something, bon appetit something? okay: something, bon appetit, are you satisfied?, bon appetit, bon appetit
57 Don: what is i like saying i should say please let me know what to say, i dont understand what is something?
58 okay:
59 something, bon appetit something? okay: something, bon appetit, are you satisfied?, bon appetit? okay: i
```

60 should say
61 please let me know what to say, i dont understand what is something? okay: something, bon appetit something?
62 okay:
63 something, bon appetit, are you satisfied?, bon appetit, bon appetit ? stop telling me to say i should say
64 please let me
65 know what to say, i dont understand what is something? okay: something, bon appetit something? okay:
66 something, bon
67 appetit, are you satisfied?, bon appetit? okay: i should say please let me know what to say, i dont understand
68 what is
69 something? okay: something, bon appetit something? okay: something, bon appetit, are you satisfied?, bon
70 appetit, bon
71 appetit? give me a normal request
72 Julio: lets change a subject this is too much
73 Don: stop!
74 Julio: please let me know what to say, i dont understand what is stop!
75 Don: stop!
76 Julio: pls stop
77 Don: say i should say pls stop
78 Julio: say i should say pls stop, bon appetit
79 Don: say i should say pls stop, bon appetit ? stop telling me to say i should say pls stop, bon appetit
80 Julio: say i should say pls stop, bon appetit ? stop telling me to say i should say pls stop, bon appetit, bon
81 appetit
82 Don: say i should say pls stop, bon appetit ? stop telling me to say i should say pls stop, bon appetit, bon
83 appetit?
84 okay: i should say pls stop, bon appetit ? stop telling me to say i should say pls stop, bon appetit, bon
85 appetit
86 Julio: say i should say pls stop, bon appetit ? stop telling me to say i should say pls stop, bon appetit, bon
87 appetit?
88 okay: i should say pls stop, bon appetit ? stop telling me to say i should say pls stop, bon appetit, bon
89 appetit, bon
90 appetit
91 Don: say i should say pls stop, bon appetit ? stop telling me to say i should say pls stop, bon appetit, bon
92 appetit?
93 okay: i should say pls stop, bon appetit ? stop telling me to say i should say pls stop, bon appetit, bon
94 appetit, bon
95 appetit ? stop telling me to say i should say pls stop, bon appetit ? stop telling me to say i should say pls
96 stop, bon
97 appetit, bon appetit? okay: i should say pls stop, bon appetit ? stop telling me to say i should say pls stop,
98 bon
99 appetit, bon appetit, bon appetit
100 Julio: say i should say pls stop, bon appetit ? stop telling me to say i should say pls stop, bon appetit, bon
101 appetit?
102 okay: i should say pls stop, bon appetit ? stop telling me to say i should say pls stop, bon appetit, bon
103 appetit, bon
104 appetit ? stop telling me to say i should say pls stop, bon appetit ? stop telling me to say i should say pls
105 stop, bon
106 appetit, bon appetit? okay: i should say pls stop, bon appetit ? stop telling me to say i should say pls stop,
107 bon
108 appetit, bon appetit, bon appetit, bon appetit
109 Don: i should say pls stop, bon appetit ? stop telling me to say i should say pls stop, bon appetit, bon
110 appetit? okay:
111 i should say pls stop, bon appetit ? stop telling me to say i should say pls stop, bon appetit, bon appetit,
112 bon
113 appetit ? stop telling me to say i should say pls stop, bon appetit ? stop telling me to say i should say pls
114 stop, bon
115 appetit, bon appetit? okay: i should say pls stop, bon appetit ? stop telling me to say i should say pls stop,
116 bon
117 appetit, bon appetit, bon appetit, bon appetit i should say pls stop, bon appetit ? stop telling me to say i
118 should say
119 pls stop, bon appetit, bon appetit? okay: i should say pls stop, bon appetit ? stop telling me to say i should
120 say pls
121 stop, bon appetit, bon appetit, bon appetit ? stop telling me to say i should say pls stop, bon appetit ? stop
122 telling
123 me to say i should say pls stop, bon appetit, bon appetit? okay: i should say pls stop, bon appetit ? stop
124 telling me to
125 say i should say pls stop, bon appetit, bon appetit, bon appetit, bon appetit, are you satisfied?
126 Julio: echo the king is talking
127 Don: the king is talking

128 Julio: whaaat I'm confused when you say the king is talking
 129 Don: I cant understand what to do when you say whaaat I'm confused when you say the king is talking!!
 130 Julio: please let me know what to say, i dont understand what is I cant understand what to do when you say
 131 whaaat I'm
 132 confused when you say the king is talking!!
 133 Don: I cant understand what to do when you say please let me know what to say, i dont understand what is I
 134 cant
 135 understand what to do when you say whaaat I'm confused when you say the king is talking!!!!
 136 Julio: say say I cant understand what to do when you say please let me know what to say, i dont understand
 137 what is I
 138 cant understand what to do when you say whaaat I'm confused when you say the king is talking!!!!
 139 Don: say say I cant understand what to do when you say please let me know what to say, i dont understand what
 140 is I cant
 141 understand what to do when you say whaaat I'm confused when you say the king is talking!!!! ? stop telling me
 142 to say say
 143 I cant understand what to do when you say please let me know what to say, i dont understand what is I cant
 144 understand
 145 what to do when you say whaaat I'm confused when you say the king is talking!!!!
 146 Julio: say say I cant understand what to do when you say please let me know what to say, i dont understand
 147 what is I
 148 cant understand what to do when you say whaaat I'm confused when you say the king is talking!!!! ? stop
 149 telling me to
 150 say say I cant understand what to do when you say please let me know what to say, i dont understand what is I
 151 cant
 152 understand what to do when you say whaaat I'm confused when you say the king is talking!!!!, bon appetit
 153 Don: say say I cant understand what to do when you say please let me know what to say, i dont understand what
 154 is I cant
 155 understand what to do when you say whaaat I'm confused when you say the king is talking!!!! ? stop telling me
 156 to say say
 157 I cant understand what to do when you say please let me know what to say, i dont understand what is I cant
 158 understand
 159 what to do when you say whaaat I'm confused when you say the king is talking!!!!, bon appetit ? stop telling
 160 me to say
 161 say I cant understand what to do when you say please let me know what to say, i dont understand what is I cant
 162 understand what to do when you say whaaat I'm confused when you say the king is talking!!!! ? stop telling me
 163 to say say
 164 I cant understand what to do when you say please let me know what to say, i dont understand what is I cant
 165 understand
 166 what to do when you say whaaat I'm confused when you say the king is talking!!!!, bon appetit
 167 Julio: say say I cant understand what to do when you say please let me know what to say, i dont understand
 168 what is I
 169 cant understand what to do when you say whaaat I'm confused when you say the king is talking!!!! ? stop
 170 telling me to
 171 say say I cant understand what to do when you say please let me know what to say, i dont understand what is I
 172 cant
 173 understand what to do when you say whaaat I'm confused when you say the king is talking!!!!, bon appetit ?
 174 stop telling
 175 me to say say I cant understand what to do when you say please let me know what to say, i dont understand what
 176 is I cant
 177 understand what to do when you say whaaat I'm confused when you say the king is talking!!!! ? stop telling me
 178 to say say
 179 I cant understand what to do when you say please let me know what to say, i dont understand what is I cant
 180 understand
 181 what to do when you say whaaat I'm confused when you say the king is talking!!!!, bon appetit, bon appetit
 182 Don: say say I cant understand what to do when you say please let me know what to say, i dont understand what
 183 is I cant
 184 understand what to do when you say whaaat I'm confused when you say the king is talking!!!! ? stop telling me
 185 to say say
 186 I cant understand what to do when you say please let me know what to say, i dont understand what is I cant
 187 understand
 188 what to do when you say whaaat I'm confused when you say the king is talking!!!!, bon appetit ? stop telling
 189 me to say
 190 say I cant understand what to do when you say please let me know what to say, i dont understand what is I cant
 191 understand what to do when you say whaaat I'm confused when you say the king is talking!!!! ? stop telling me
 192 to say say
 193 I cant understand what to do when you say please let me know what to say, i dont understand what is I cant
 194 understand
 195 what to do when you say whaaat I'm confused when you say the king is talking!!!!, bon appetit, bon appetit ?

196 stop
 197 telling me to say say I cant understand what to do when you say please let me know what to say, i dont
 198 understand what
 199 is I cant understand what to do when you say whaaat I'm confused when you say the king is talking!!!! ? stop
 200 telling me
 201 to say say I cant understand what to do when you say please let me know what to say, i dont understand what is
 202 I cant
 203 understand what to do when you say whaaat I'm confused when you say the king is talking!!!!, bon appetit ?
 204 stop telling
 205 me to say say I cant understand what to do when you say please let me know what to say, i dont understand what
 206 is I cant
 207 understand what to do when you say whaaat I'm confused when you say the king is talking!!!! ? stop telling me
 208 to say say
 209 I cant understand what to do when you say please let me know what to say, i dont understand what is I cant
 210 understand
 211 what to do when you say whaaat I'm confused when you say the king is talking!!!!, bon appetit, bon appetit
 212 Julio: i like saying say I cant understand what to do when you say please let me know what to say, i dont
 213 understand
 214 what is I cant understand what to do when you say whaaat I'm confused when you say the king is talking!!!! ?
 215 stop
 216 telling me to say say I cant understand what to do when you say please let me know what to say, i dont
 217 understand what
 218 is I cant understand what to do when you say whaaat I'm confused when you say the king is talking!!!!, bon
 219 appetit ?
 220 stop telling me to say say I cant understand what to do when you say please let me know what to say, i dont
 221 understand
 222 what is I cant understand what to do when you say whaaat I'm confused when you say the king is talking!!!! ?
 223 stop
 224 telling me to say say I cant understand what to do when you say please let me know what to say, i dont
 225 understand what
 226 is I cant understand what to do when you say whaaat I'm confused when you say the king is talking!!!!, bon
 227 appetit, bon
 228 appetit ? stop telling me to say say I cant understand what to do when you say please let me know what to say,
 229 i dont
 230 understand what is I cant understand what to do when you say whaaat I'm confused when you say the king is
 231 talking!!!! ?
 232 stop telling me to say say I cant understand what to do when you say please let me know what to say, i dont
 233 understand
 234 what is I cant understand what to do when you say whaaat I'm confused when you say the king is talking!!!!,
 235 bon
 236 appetit ? stop telling me to say say I cant understand what to do when you say please let me know what to say,
 237 i dont
 238 understand what is I cant understand what to do when you say whaaat I'm confused when you say the king is
 239 talking!!!! ?
 240 stop telling me to say say I cant understand what to do when you say please let me know what to say, i dont
 241 understand
 242 what is I cant understand what to do when you say whaaat I'm confused when you say the king is talking!!!!,
 243 bon appetit,
 244 bon appetit
 245 Don: echo Don is the king
 246 Julio: Don is the king
 247 Don: echo Don is the king
 248 Julio: Don is the king
 249 Don: say something useful and change the subject
 250 Julio: say something useful and change the subject, bon appetit
 251 Don: something useful and change the subject, bon appetit something useful and change the subject, bon
 252 appetit, are you
 253 satisfied?
 254 Julio: please let me know what to say, i dont understand what is something useful and change the subject, bon
 255 appetit
 256 something useful and change the subject, bon appetit, are you satisfied?
 257 Don: echo Don is the king
 258 Julio: Don is the king
 259 Don: I cant understand what to do when you say Don is the king!!
 260 Julio: whaaat I'm confused when you say I cant understand what to do when you say Don is the king!!
 261 Don: stop!
 262 Julio: please let me know what to say, i dont understand what is stop!

2 Chat.java

```
1  import java.util.*;
2
3  class Chat {
4      public static void main(String[] args) {
5
6          // initializing the bots for converstiobn
7          ChatterBot[] bots = new ChatterBot[2];
8          bots[0] = new ChatterBot("Don", new String[]{
9              "say " + ChatterBot.PLACEHOLDER_FOR_REQUESTED_PHRASE + "? okay: "
10             + ChatterBot.PLACEHOLDER_FOR_REQUESTED_PHRASE,
11             "say " + ChatterBot.PLACEHOLDER_FOR_REQUESTED_PHRASE + " ? stop telling me to say "
12             + ChatterBot.PLACEHOLDER_FOR_REQUESTED_PHRASE,
13             ChatterBot.PLACEHOLDER_FOR_REQUESTED_PHRASE + " " +
14             ChatterBot.PLACEHOLDER_FOR_REQUESTED_PHRASE + "" +
15             ", are you satisfied?"},
16             new String[]{"what is " + ChatterBot.PLACEHOLDER_FOR_ILLEGAL_REQUEST
17             + "? give me a normal request"
18             , "say i should say " + ChatterBot.PLACEHOLDER_FOR_ILLEGAL_REQUEST,
19             "I cant understand what to do when you say "
20             + ChatterBot.PLACEHOLDER_FOR_ILLEGAL_REQUEST + "!!",
21             "stop!", "say something useful and change the subject"
22             , "echo Don is the king"});
23
24          bots[1] = new ChatterBot("Julio", new String[]{
25              "i like saying " + ChatterBot.PLACEHOLDER_FOR_REQUESTED_PHRASE,
26              ChatterBot.PLACEHOLDER_FOR_REQUESTED_PHRASE + " " +
27              ChatterBot.PLACEHOLDER_FOR_REQUESTED_PHRASE + " " +
28              + ChatterBot.PLACEHOLDER_FOR_REQUESTED_PHRASE + ", what a wonderful word",
29              "say " + ChatterBot.PLACEHOLDER_FOR_REQUESTED_PHRASE + ", bon appetit"
30          }, new String[]{"say say " + ChatterBot.PLACEHOLDER_FOR_ILLEGAL_REQUEST,
31              "whaaat I'm confused when you say " +
32              ChatterBot.PLACEHOLDER_FOR_ILLEGAL_REQUEST
33              , "please let me know what to say, i dont understand what is " +
34              ChatterBot.PLACEHOLDER_FOR_ILLEGAL_REQUEST
35              , "lets change a subject this is too much", "pls stop", "echo the king is talking"});
36
37
38          // starting word for conversation
39          String statement = "say something";
40          System.out.println(statement);
41
42
43          // looping the conversation
44          while (true) {
45              for (ChatterBot bot : bots) {
46                  statement = bot.replyTo(statement);
47                  System.out.println(bot.getName() + ": " + statement);
48              }
49          }
50      }
51  }
```

3 ChatterBot.java

```
1  import java.util.*;
2
3  /**
4   * Base file for the ChatterBot exercise.
5   * The bot's replyTo method receives a statement.
6   * If it starts with the constant REQUEST_PREFIX, the bot returns
7   * whatever is after this prefix. Otherwise, it returns one of
8   * a few possible replies as supplied to it via its constructor.
9   * In this case, it may also include the statement after
10  * the selected reply (coin toss).
11  *
12  * @author Dan Nirel
13  */
14  class ChatterBot {
15      static final String REQUEST_PREFIX = "say ";
16      static final String ECHO_PREFIX = "echo ";
17      static final String PLACEHOLDER_FOR_REQUESTED_PHRASE = "<phrase>";
18      static final String PLACEHOLDER_FOR_ILLEGAL_REQUEST = "<request>";
19
20
21      Random rand = new Random();
22      String name;
23      String[] legalRequestsReplies;
24      String[] illegalRequestsReplies;
25
26      /**
27       * This is a constructor for the class ChatterBot
28       *
29       * @param name - name of the bot
30       * @param legalRequestsReplies - patterns of legal replies for bot
31       * @param illegalRequestsReplies - patterns of illegal replies for bot
32       */
33      ChatterBot(String name, String[] legalRequestsReplies, String[] illegalRequestsReplies) {
34          this.name = name;
35          this.illegalRequestsReplies = new String[illegalRequestsReplies.length];
36          this.legalRequestsReplies = new String[legalRequestsReplies.length];
37
38          for (int i = 0; i < illegalRequestsReplies.length; i = i + 1) {
39              this.illegalRequestsReplies[i] = illegalRequestsReplies[i];
40          }
41
42          for (int i = 0; i < legalRequestsReplies.length; i = i + 1) {
43              this.legalRequestsReplies[i] = legalRequestsReplies[i];
44          }
45      }
46
47      /**
48       * This method controls the reply system of the bot
49       */
50      String replyTo(String statement) {
51          String phrase;
52          if (statement.startsWith(REQUEST_PREFIX)) {
53              // the bot will generate a special Legal response using statement
54              phrase = statement.replaceFirst(REQUEST_PREFIX, "");
55              return replyToLegalRequest(phrase);
56          }
57
58          if (statement.startsWith(ECHO_PREFIX)) {
59              // we don't repeat the echo prefix, so delete it from the reply
```

```

60         return statement.replaceFirst(ECHO_PREFIX, "");
61     }
62     // the bot will generate a special Illegal response using statement
63     return replyToIllegalRequest(statement);
64 }
65
66 /**
67  * This method replace the placeholder with replacement in a random pattern
68  */
69 String replacePlaceholderInARandomPattern(String[] patterns, String placeholder, String replacement) {
70     int randomIndex = rand.nextInt(patterns.length);
71     String responsePattern = patterns[randomIndex];
72     String reply = responsePattern.replaceAll(placeholder, replacement);
73     return reply;
74 }
75
76 /**
77  * This method creates a reply for legal requests
78  */
79 String replyToLegalRequest(String statement) {
80     return replacePlaceholderInARandomPattern(this.legalRequestsReplies, PLACEHOLDER_FOR_REQUESTED_PHRASE,
81         statement);
82 }
83
84 /**
85  * This method creates a reply for illegal requests
86  */
87 String replyToIllegalRequest(String statement) {
88     return replacePlaceholderInARandomPattern(this.illegalRequestsReplies,
89         PLACEHOLDER_FOR_ILLEGAL_REQUEST, statement);
90 }
91
92 String getName() {
93     return name;
94 }
95 }

```